Nederlandse norm

NEN-EN 1005-2+A1

(en)

Safety of machinery - Human physical performance - Part 2: Manual handling of machinery and component parts of machinery

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Vervangt NEN-EN 1005-2:2003

ICS 13.110-30; 13.180

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Nederlands voorwoord

Voor de in deze norm vermelde normatieve verwijzingen bestaan in Nederland de volgende equivalenten:

vermelde norm	Nederlandse norm	titel
EN 292-2:1991	-	-
EN 614-1	NEN-EN 614-1	Veiligheid van machines - Ergonomische ontwerpprincipes - Deel 1: Terminologie en algemene principes (en)
EN 1005-1:2001	NEN-EN 1005-1:2001	Veiligheid van machines - Menselijke fysieke belasting - Deel 1: Termen en definities (en)
EN 1050	-	-
EN 1070:1998	-	-

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 1005-2:2003+A1

October 2008

ICS 13.110; 13.180

Supersedes EN 1005-2:2003

English Version

Safety of machinery - Human physical performance - Part 2: Manual handling of machinery and component parts of machinery

Sécurité des machines - Performance physique humaine - Partie 2: Manutention manuelle de machines et d'éléments de machines

Sicherheit von Maschinen - Menschliche körperliche Leistung - Teil 2: Manuelle Handhabung von Gegenständen in Verbindung mit Maschinen und Maschinenteilen

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Foreword

This document (EN 1005-2:2003+A1:2008) has been prepared by Technical Committee CEN /TC 122, "Ergonomics", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by April 2009, and conflicting national standards shall be withdrawn at the latest by December 2009.

This document includes Amendment 1, approved by CEN on 2008-08-18.

This document supersedes EN 1005-2:2003.

The start and finish of text introduced or altered by amendment is indicated in the text by tags ▌A1⟩ ⟨A1▌.

This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association and supports essential requirements of EU Directive(s).

▌A1⟩ For relationship with EU Directive(s), see informative Annexes ZA and ZB, which are integral parts of this document. ⟨A1▌

EN 1005 consists of the following parts, under the general title "Safety of machinery - Human physical performance":

— Part 1: Terms and definitions;

— Part 2: Manual handling of machinery and component parts of machinery;

— Part 3: Recommended force limits for machinery operation;

— Part 4[1)]: Evaluation of working postures and movements in relation to machinery;

— Part 5[1)]: Risk assessment for repetitive handling at high frequency.

Annexes A, B and C are for informative.

This document includes a Bibliography.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1) This European Standard is under preparation by CEN/TC 122/WG 4 "Biomechanics".

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Introduction

When designing and constructing machinery, the manufacturer should meet the essential requirements for safety and health set out in the Machinery Directive. The manufacturer should assess the hazards of the machine and consider these hazards related to the life cycle of the machine.

This European Standard is one of several ergonomic standards for the safety of machinery. EN 614-1 describes the principles to be applied by designers in order to accommodate ergonomic factors.

This standard has been prepared to be a harmonised standard in the sense of the Machinery Directive and associated EFTA regulations.

This European Standard is a type B standard as stated in EN 1070.

The provisions of this document can be supplemented or modified by a type C standard.

NOTE For machines which are covered by the scope of a type C standard and which have been designed and built according to the provisions of that standard, the provisions of that type C standard take precedence over the provisions of this type B standard.

Manual handling of loads can lead to a high risk of injury to the musculoskeletal system if the loads to be handled are too heavy, and/or handled at high frequencies for long durations and/or in awkward postures. Disorders of the musculoskeletal system are of a common occurrence throughout Europe. Manually applied effort is often required by operators working with machines for their intended purpose. Risks exist if the design of the machinery is not in accordance with ergonomic design principles. When designing and constructing machinery where manual handling is required, this standard provides relevant data for working posture, load, frequency and duration. The design criteria given in this standard can be used by the designer when making risk assessments.

This standard requires machinery designers to adopt a three stage approach to:

a) avoid manual handling activities wherever possible;

b) utilise technical aids;

c) further reduce the inherent level of risk by optimising handling activities.

For machines and their component parts which cannot be moved or transported by hand, see 4.2 in EN 292-2:1991.

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1 Scope

This European Standard specifies ergonomic recommendations for the design of machinery involving manual handling of machinery and component parts of machinery, including tools linked to the machine, in professional and domestic applications.

This European Standard applies to the manual handling of machinery, component parts of machinery and objects processed by the machine (input/output) of 3 kg or more, for carrying less than 2 m. Objects of less than 3 kg are dealt with in prEN 1005-5[1)]. The standard provides data for ergonomic design and risk assessment concerning lifting, lowering and carrying in relation to the assembly/erection, transport and commissioning (assembly, installation, adjustment), operation, fault finding, maintenance, setting, teaching or process changeover and decommissioning, disposal and dismantling of machinery.

This standard provides current data on the general population and certain sub-populations (clarified in annex A).

This part of the standard does not cover the holding of objects (without walking), pushing or pulling of objects, hand-held machines, or handling while seated.

This document is not applicable to specify the machinery which are manufactured before the date of publication of this document by CEN.

2 Normative references

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text, and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

EN 292-2:1991, *Safety of machinery – Basic concepts, general principles for design – Part 2: Technical principles and specifications.*

EN 614-1, *Safety of machinery – Ergonomic design principles – Part 1: Terminology and general principles.*

EN 1005-1:2001*, Safety of machinery – Human physical performance – Part 1: Terms and definitions.*

EN 1050*, Safety of machinery – Principles for risk assessment.*

EN 1070:1998, *Safety of machinery – Terminology.*

3 Terms and definitions

For the purposes of this European Standard, the terms and definitions given in EN 1005-1:2001 and EN 1070:1998 apply.

4 Recommendations for the design of machinery and component parts where objects are lifted, lowered and carried

4.1 General principles

In order to minimise the risks to the health and safety of the operator, when lifting, lowering and carrying the machine or component parts, the designer/manufacturer of the machine shall:

a) establish whether or not a hazard exists when manual handling activities are performed in relation to the intended use of the machine. If a hazard exists, the risk assessment model should be applied. (see 4.3.1, 4.3.2 and 4.3.3);

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b) remove the hazard by excluding the need for manual handling activities (lifting, lowering and carrying) for the machinery or machinery-linked objects (see 4.2.1). If this is not possible, provide technical aids, additional to the system (see 4.2.2), and/or design/redesign the machinery and machinery-linked objects according to ergonomic principles (see 4.3);

c) provide technical specifications and instructions so that machinery is used appropriately and in the intended manner covered by the risk assessment, technical specifications and instructions for use.

In all cases, the designer shall provide instructions concerning construction, transport and commissioning, use and decommissioning of the machinery (see 4.4) to prevent risks for safety and health.

Technical aids for manual handling should preferably be incorporated into the machinery design and the machine.

Health, safety and productivity are most likely to be optimised if an ergonomic approach is used in designing the system of work as a whole.

4.2 Recommendations for the design of machinery

4.2.1 Systems without manual handling

The best way to eliminate manual handling risks is to remove the need to handle manually. In general, those designing new systems of work or installing new systems of machinery should consider introducing an integrated handling system that fully utilises powered handling, rather than a manual system.

4.2.2 Manual handling with technical aids

Designers should ascertain whether suitable aids and equipment can be installed, particularly if the existing practice involves frequent manual handling of objects and manual handling of objects weighing more than the reference mass (see Table 1), or the precise movement of objects. Appropriate technical aids diminish or eliminate the risk of musculoskeletal injury but may induce risks, for example, by the need for maintenance work. They may also have implications for particular handling methods at subsequent points in the handling chain. Handling aids should be compatible with the rest of the work system, including layout and access routes. They should be effective for the full range of operational conditions likely to be encountered. In employing technical aids, more space could be required for access to allow appropriate working postures.

When designing systems including manual handling follow the recommendations given in 4.3.2.

When considering the limits for manually pushing and pulling of technical aids, reference should be made to EN 1005-3. When considering working postures, reference should be made to prEN 1005-4[1)].

4.3 Risk assessment and recommendations for the design of machinery and component parts involving manual handling

4.3.1 Approach to risk assessment

To control the risks to an individual's health and safety, it is necessary for the designer to identify relevant hazards and to carry out an appropriate risk assessment. For general requirements of risk assessment, refer to EN 1050. Manual handling risks shall be reduced by applying ergonomic principles at the design stage of the machinery, see EN 614-1.

The risk assessment model presented here involves 3 methods. These methods have the same basis, but differ in their complexity of application.

The first method is a quick screening method. Method 2, an easy to handle method, shall be applied if the screening method indicates risks. Some additional risk factors can be taken into account in method 2. Method 3 is an extended assessment method, which assesses risks in a more thorough way and is supplemented by additional risk factors not presented in methods 1 and 2. All three methods have different levels of complexity. The most efficient approach is to begin the risk assessment by applying method 1 (the most simple one) and use methods 2 and/or 3 only if the assumptions and/or operational situations identified in method 1 are not met.

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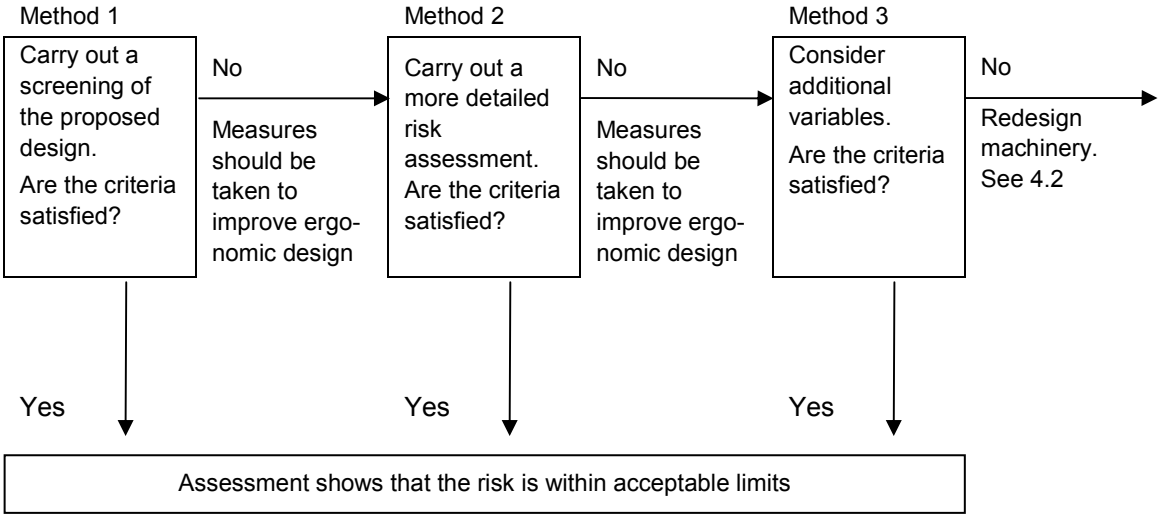

NOTE It is recommended to consider further steps to reduce risk factors to their lowest possible level.

Figure 1 — Flowchart identifying the step-wise approach to assessment

4.3.2 Hazard identification, risk estimation, risk evaluation and recommendations for risk reduction by design

In this paragraph recommendations for the design of a manual handling situation at a low risk level are given. The information is based on the state of the art of ergonomics. If these criteria are not met a risk assessment should be done.

The following aspects (see 4.3.2.1 to 4.3.2.3) shall be taken into account when assessing and reducing the risks.

4.3.2.1 Objects

4.3.2.1.1 Mass

The mass of the object (machines, machinery parts, input/output) includes everything connected with it such as packing, batteries, full magazines, etc. Technical aids needed for the manual handling activities are also included.

When designing machinery or components, the unit mass can be a hazard if an object is to be handled manually. To define the safe maximum limit for the mass refer to 4.3.3.

4.3.2.1.2 Mass distribution/stability

The object ‘s centre of gravity is determined by the distribution of mass. Wherever possible, the centre of gravity of the object should be within the object, distributed equally between both hands and positioned as close as possible to the body. While handling, movement of the object's centre of gravity should be constrained. When this is not possible (e.g. liquid), appropriate information should be marked on the object.

4.3.2.1.3 Size

Objects should be designed to be as compact as possible. When objects are gripped by both hands, the width of the object should not exceed shoulder width (approximately 60 cm) and the depth of the object should not exceed 50 cm (recommended to be 35 cm or less) in order to keep the load close to the body. The height of the object should be such that it does not obscure a person's visibility. If handling of the object needs one hand underneath and one hand above the object, the object should be redesigned.

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4.3.2.1.4 Grip/handles

The surface properties of objects (including packaging materials of new component parts) should be suitable for handling. An object should be easy to grasp and hold and, therefore, should normally be equipped with handles or suitable cut-outs.

The placement of the handles should be consistent with the centre of gravity of the object and the type of action that is carried out, supporting to adopt the best working postures and movement during lifting and carrying. If the handle is supposed to be horizontal, this should be done by the machine. Handles should not have sharp edges or contain a risk of compressing the fingers. The shape of the handle should permit a hook-grip or power-grip with a neutral hand-arm posture.

The diameter of the handle should be between 2 cm and 4 cm.

The width of handle/cut-out should be a minimum of 12,5 cm to ensure clearance for a gloved hand, with 7 cm room above the fingers. The optimum shape of the handle should be cylindrical or elliptical. To assess the handgrip (coupling) go to 4.3.3.2. or 4.3.3.3.

4.3.2.2 Operation-machine interface

4.3.2.2.1 Horizontal location and distance

Obstruction between the object and the body, such that the object cannot be lifted, lowered or carried close to the body, is a hazard. The horizontal distance between the mid-point of the ankles and of the hands should be less than 25 cm. If the horizontal distance is greater than this value, a risk assessment should be carried out in accordance with 4.3.3.

4.3.2.2.2 Vertical location and displacement

The vertical grip position (measured from the knuckle of the middle finger to the ground) should be adjustable between 60 cm and 90 cm. Vertical displacement of the object should not exceed 25 cm. If the vertical distance is lower or higher, and/or the vertical displacement exceeds 25 cm, then a risk assessment should be carried out in accordance with 4.3.3.

4.3.2.2.3 Frequency of operation

Machinery should be designed so that manual handling at high frequency is avoided. It should be possible to regulate the frequency of the machine in relation to manual handling. A machinery design should, wherever possible, allow some degree of autonomy. To assess the risks arising from a combination of the frequency and mass, refer to 4.3.3.

4.3.2.2.4 Working postures

The designer should avoid awkward postures (e.g. twisting and bending, lifting from the ground) and prolonged activities which lead to body fatigue during manual handling, for every stage of machinery-operation (including assembly/erection, transport and commissioning, use and decommissioning). Occasional changes in posture should be provided (see prEN 1005-4).

4.3.2.2.5 Manual carrying of loads

In general, machines should be designed so that manual carrying is avoided. Where this is not possible, the maximum manual carrying distance should be as low as possible (less than 2 m).

4.3.2.2.6 One-handed handling

If an object is handled with one hand only it may be a hazard. If one-hand handling cannot be avoided a risk assessment should be carried out (see 4.3.3.3).

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4.3.2.2.7 Handling by two persons

Handling by two (or more) persons can be used to reduce the load on one operator, but it also creates additional hazards because of difficulties in co-ordinating the movements and force exertions between the two (or more) people performing the lift together. Ergonomic design shall eliminate the need for using two (or more) people for lifts except in occasionally occurring special cases, for which a risk assessment should be carried out (see 4.3.3.3).

4.3.2.2.8 Handling including rotation of object

Manual handling of objects by turning around its axis should be avoided. Redesign is recommended by which machinery or technical aids take over the handling.

4.3.2.2.9 Additional physical demands

Additional physical demands to the manual handling task may imply new hazards. If they cannot be avoided, a risk assessment should be carried out (see 4.3.3.3).

4.3.2.2.10 Coupling

Unfavourable coupling conditions between hands and the objects handled or between the feet and floor may result in hazardous situations. If they cannot be avoided, a risk assessment should be carried out (see 4.3.3.3).

4.3.2.3 Environmental factors

Environmental factors can create hazards and can impose additional risk.

If they cannot be avoided, qualified personnel should make a risk assessment. Environmental factors include vibration, climate, thermal conditions (see EN ISO 7730), illumination, slippery ground, noise and chemical agents (see ENV 26385).

4.3.3 Risk assessment model

This risk assessment model consists of three methods. Method 1 is a quick screening method. Method 2 shall be applied if the screening method indicates risks. Some additional risk factors can be taken into account in method 2. Method 3 is an extended assessment method which assesses risks in a more thorough way and is supplemented by additional risk factors not presented in methods 1 or 2. All three methods have different levels of complexity. The most efficient approach is to begin the risk assessment by applying method 1 (the simplest procedure) and use methods 2 or 3 only if the assumptions or operational situations identified in method 1 are not met. Each method requires three steps to be carried out:

— Step 1: consider the reference mass in relation to the intended user population (see Table 1);

— Step 2: carry out the risk assessment according to the worksheet, see 4.3.3.1 to 4.3.3.3;

— Step 3: identify the action required:

  — no action, if risk level is tolerable;

  — redesign or;

  — use a more complex assessment method.

It is free to the designer to redesign already after Method 1, step 3 and not to go to Method 2. It is not a complete iteration but a more detailed risk assessment in Method 2 and Method 3, based on same methodology.

For practical application see annex C.

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Table 1 — Reference mass (M_{ref}) taking into consideration the intended user population

Field of application	M_{ref} [kg]	Percentage of			Population group	
		F and M	F_{emales}	M_{ales}		
Domestic use[a]	5	Data not available			Children and the elderly	Total population
	10	99	99	99	General domestic population	
Professional use (general)[b]	15	95	90	99	General working population, including the young and old	General working population
	25	85	70	90	Adult working population	
Professional use (exceptional)[c]	30 35 40	Data not available			Special working population	Special working population

[a] When designing a machine for domestic use, 10 kg should be used as a general reference mass in the risk assessment. If children and elderly are included in the intended user population, the reference mass should be lowered to 5 kg.

[b] When designing a machine for professional use, a reference mass of 25 kg should not be exceeded in general.

[c] While every effort should be made to avoid manual handling activities or reduce the risks to the lowest possible level, there may be exceptional circumstances where the reference mass might exceed 25 kg (e.g. where technological developments or interventions are not sufficiently advanced). Under these special conditions other measures have to be taken to control the risk according to EN 614-1 (e.g. technical aids, instructions and / or special training for the intended operator group).

4.3.3.1 Method 1: Screening by means of critical values

In this method the designer may choose one of three operational situations which is appropriate to the design. By using this method a quick screening can be carried out. Limiting condition is that assumptions for handling operation are fulfilled.

4.3.3.1.1 Consider the reference mass (step 1)

Identify the intended user population and select the reference mass (M_{ref}) according to the intended user population (Table 1).

4.3.3.1.2 Carry out the risk assessment (step 2)

Identify (tick) whether the handling operation meets the following criteria:

— two-handed operation only;

— unrestricted standing posture and movements;

— handling by one person only;

— smooth lifting;

— good coupling between the hands and the objects handled;

— good coupling between the feet and floor;

— manual handling activities, other than lifting, are minimal;

— the objects to be lifted are not very cold[2)] , hot or contaminated;

2) see prEN ISO 13732-3 'Ergonomics of the thermal environment – Touching of cold surfaces – Part 3: Ergonomics data and guidance for application (ISO/DIS 13732-3:2002)'

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— moderate ambient thermal environment.

If one or more of these criteria are not met, method 2 applies. If all criteria are met, then select one of the following critical variables. These apply to a work shift of 8 h or less.

4.3.3.1.2.1 Critical variables

a) Critical mass (case 1)

— the load handled does not exceed 70 % of the reference mass selected from Table 1;

— vertical displacement of the load is less than or equal 25 cm and does not occur below hip or above shoulder height;

— the trunk is upright and not rotated;

— the load is kept close to the body;

— the frequency of lifts is equal to or less than $3{,}33 \times 10^{-3}$ Hz (1 lift per 5 min).

b) Critical vertical mass displacement (case 2)

— the load handled does not exceed 60 % of the reference mass selected from Table 1;

— the vertical displacement does not occur above shoulder height or below knee height;

— the trunk is upright and not rotated;

— the load is kept close to the body;

— the frequency of lifts is equal or less than $3{,}33 \times 10^{-3}$ Hz (1 lift per 5 min).

c) Critical frequency (case 3)

— the load handled does not exceed 30 % of the mass selected from Table 1;

— vertical displacement of the load is less than or equal 25 cm and does not occur below hip or above shoulder height;

— the frequency of lifts is equal or less than 0,08 Hz (5 lifts per min);

— the trunk is upright and not rotated;

— the load is kept close to the body

or

— the load handled does not exceed 50 % of the reference mass selected from Table 1;

— vertical displacement of the load is less than or equal 25 cm and does not occur below hip or above shoulder height;

— the frequency of lifts is equal or less than 0,04 (2,5 lifts per min);

— the trunk is upright and not rotated;

— the load is kept close to the body.

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4.3.3.1.3 Select the action required (step 3)

If the design fits one of the operational situations (cases 1 to 3) described above, the risk assessment has been carried out successfully.

If none of the operational situations are satisfied, or any of the criteria specified in step 2 are not met, either

— consider modifying or redesign the machinery or

— use a more detailed assessment procedure to identify critical risk factors (method 2).

4.3.3.2 Method 2: Estimation by tables

4.3.3.2.1 Consider the reference mass (step 1)

Identify the intended user population and select the reference mass (M_{ref}) according to the intended user population (Table 1).

4.3.3.2.2 Carry out the risk assessment (step 2)

Identify (tick) whether the handling operation meets the following criteria:

— two-handed operation only;

— unrestricted standing posture and movements;

— handling by one person only;

— smooth lifting;

— good coupling between the feet and floor;

— manual handling activities, other than lifting are minimal;

— the objects to be lifted are not very cold, hot or contaminated;

— moderate ambient thermal environment.

If one or more of these criteria are not met, refer to method 3.

If all criteria are met, then determine the level of risk by calculating the recommended mass limit (R_{ML2}) using the multipliers provided in Table 2 and then the risk index (R_I) as follows:

$$R_I = \frac{\text{actual mass}}{R_{ML2}}$$

4.3.3.2.3 Select the action required (step 3)

— If $R_I \leq 0{,}85$, the risk may be regarded as tolerable (green).

— $0{,}85 < R_I < 1{,}0$ indicates that significant risk exists (yellow). It is recommended that:

— method 3 is applied in order to identify how the risk may be reduced or

— the machinery be either redesigned or

— ensure that the risk is tolerable.

$R_I \geq 1{,}0$ means that redesign is necessary. The design can be improved by changing the situations that lead to low multipliers (red).

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Calculate the recommended mass limit (R_{ML2}) for method 2 as follows:

$R_{ML2} = M_{ref} \times V_M \times D_M \times H_M \times A_M \times C_M \times F_M$,

whereas the reference mass (M_{ref}) is indicated in kg.

Table 2 — Multipliers for the calculation of the recommended mass limit (R_{ML2})

Vertical multiplier (V_M)							
Vertical location[3] [cm]	0	25	50	75	100	130	> 175
Multiplier	0,78	0,85	0,93	1,00	0,93	0,84	0,00
Distance multiplier (D_M)							
Vertical displacement[3] [cm]	25	30	40	50	70	100	> 175
Multiplier	1,00	0,97	0,93	0,91	0,88	0,87	0,00
Horizontal multiplier (H_M)							
Horizontal location[3] [cm]	25	30	40	50	56	60	> 63
Multiplier	1,00	0,83	0,63	0,50	0,45	0,42	0,00
Asymmetric multiplier (A_M)							
Angle of asymmetry[3] [°]	0	30	60	90	120	135	> 135
Multiplier	1,00	0,90	0,81	0,71	0,62	0,57	0,00

Coupling multiplier (C_M)			
Quality of grip	Good	Fair	Poor
Description	see annex C	see annex C	see annex C
Multiplier	1,00	0,95	0,90

Frequency multiplier (F_M)[4]							
	Frequency						
Hz	0,0033	0,0166	0,0666	0,1000	0,1500	0,2000	> 0,2500
lifts/min	0,20	1	4	6	9	12	> 15
Duration (d) d ≤ 1h	1,00	0,94	0,84	0,75	0,52	0,37	0,00
1h < d ≤ 2h	0,95	0,88	0,72	0,50	0,30	0,00	0,00
2h < d ≤ 8h	0,85	0,75	0,45	0,27	0,00	0,00	0,00

3) For definition see EN 1005-1.

4) For high frequencies see prEN 1005-5.

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4.3.3.3 Method 3: Calculation by formula

4.3.3.3.1 Consider the reference mass (step 1)

Identify the intended user population and

select the reference mass (M_{ref}) according to the intended user population (Table 1)

4.3.3.3.2 Carry out the risk assessment (step 2)

Identify (tick) whether the handling operation meets the following criteria:

— unrestricted standing posture and movements;

— smooth lifting;

— good coupling between the feet and floor;

— the objects to be lifted are not very cold, hot or contaminated;

— moderate ambient thermal environment.

If one or more of these criteria are not met, ensure that the risk is tolerable (see EN 614-1).

If all criteria are met, then carry out the risk estimation by calculation of the recommended mass limit (R_{ML}).

4.3.3.3.2.1 Risk estimation by calculation of the recommended mass limit (R_{ML2})

a) Case 1

Use the calculated recommended mass limit of method 2 (R_{ML2}) in the equation:

$$R_{ML} = R_{ML2} \times O_M \times P_M \times A_T$$

where:

R_{ML2} is recommended mass limit (method 2);

O_M is one handed operation, if true O_M = 0,6; otherwise O_M = 1,0;

P_M is two person operation, if true P_M = 0,85; otherwise P_M = 1,0;

A_T is the additional task multiplier, if true A_T= 0,8; otherwise A_T = 1,0.

b) Case 2

If in method 2 there are no appropriate values for vertical multiplier (V_M), distance multiplier (D_M), horizontal multiplier (H_M), asymmetric multiplier (A_M), coupling multiplier (C_M) or frequency multiplier (F_M), select a reference mass (M_{ref}) for the intended user population by referring to

Table 1 and use the following equation:

$$R_{ML} = M_{ref} \times V_M \times D_M \times H_M \times A_M \times C_M \times F_M \times O_M \times P_M \times A_T$$

the following applies:

$V_M = 1 - 0{,}003\,|V - 75|$ if $V < 0$ cm, $V_M = 0{,}78$ if $V > 175$ cm, $V_M = 0$

$D_M = 0{,}82 + 4{,}5/D$ if $D < 25$ cm, $D_M = 1$ if $D > 175$ cm, $D_M = 0$

$H_M = 25/H$ if $H < 25$ cm, $H_M = 1$ if $H > 63$ cm, $H_M = 0$

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$A_M = 1 - (0{,}0032A)$ if $A > 135°$, $A_M = 0$

where:

M_{ref} is the reference mass from Table 1, in kg;

V is the vertical location, in cm;

D is the vertical displacement, in cm;

H is the horizontal location, in cm;

A is the asymmetric angle, in degrees;

C_M is the coupling multiplier from Table 2;

F_M is the frequency multiplier from Table 3.

O_M is the one hand multiplier, if true $O_M = 0{,}6$ otherwise $O_M = 1{,}0$;

P_M is the two person multiplier, if true $P_M = 0{,}85$ otherwise $P_M = 1{,}0$;

A_T is the additional task multiplier, if true = 0,8 otherwise $A_T = 1{,}0$;

Calculate the risk index (R_I) as follows:

$$R_I = \frac{\text{actual mass}}{R_{ML}}$$

4.3.3.3.3 Select the action required (step 3)

— If $R_I \leq 0{,}85$, the risk may be regarded as tolerable.

— $0{,}85 < R_I < 1{,}0$ indicates that significant risk exists. It is recommended:

  — to redesign the machinery or

  — to ensure that the risk is tolerable.

— $R_I \geq 1{,}0$ means that redesign is necessary. The design can be improved by changing the situations that lead to low multipliers.

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Table 3 — Frequency multiplier (F_M)

Frequency		Frequency multiplier (F_M)					
		Work duration (d)					
		2 h < d ≤ 8 h		1 h < d ≤ 2 h		d ≤ 1 h	
Hz	lifts/min	V [a] < 75 cm	V [a] ≥ 75 cm	V [a] < 75 cm	V [a] ≥ 75 cm	V [a] < 75 cm	V [a] ≥ 75 cm
≤ 0,00333	≤ 0,2	0,85	0,85	0,95	0,95	1,00	1,00
0,00833	0,5	0,81	0,81	0,92	0,92	0,97	0,97
0,01666	1	0,75	0,75	0,88	0,88	0,94	0,94
0,03333	2	0,65	0,65	0,84	0,84	0,91	0,91
0,05000	3	0,55	0,55	0,79	0,79	0,88	0,88
0,06666	4	0,45	0,45	0,72	0,72	0,84	0,84
0,08333	5	0,35	0,35	0,60	0,60	0,80	0,80
0,10000	6	0,27	0,27	0,50	0,50	0,75	0,75
0,11666	7	0,22	0,22	0,42	0,42	0,70	0,70
0,13333	8	0,18	0,18	0,35	0,35	0,60	0,60
0,15000	9	0,00	0,15	0,30	0,30	0,52	0,52
0,16666	10	0,00	0,13	0,26	0,26	0,45	0,45
0,18333	11	0,00	0,00	0,00	0,23	0,41	0,41
0,20000	12	0,00	0,00	0,00	0,21	0,37	0,37
0,21666	13	0,00	0,00	0,00	0,00	0,00	0,34
0,23333	14	0,00	0,00	0,00	0,00	0,00	0,31
0,25000	15	0,00	0,00	0,00	0,00	0,00	0,28
>0,2500	> 15	0,00	0,00	0,00	0,00	0,00	0,00
[a] V is the vertical location							

4.4 Information for use

The designer should provide working instructions covering all phases of a machine's life (construction, transport and commissioning, use and decommissioning). These instructions should take into account the level of training of the operators.

For general aspects included in the working instructions see EN 292-2:1991, clause 5 and 1.7.4 of annex A.

The instructions (related to manual handling) should include:

a) General information

— purpose for intended use;

— information on foreseeable use;

— risks and residual risks;

— adequate warning of inherent risks;

— preventive measures to be taken by the user.

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b) Intended use of the machinery

— space needed;

— storage and floor surface conditions;

— conditions to diminish discomfort, fatigue and stress;

— housekeeping;

— disposal of waste.

c) Object mass

— maximum object mass;

— consequences of handling the mass.

d) Technical equipment

— when used;

— requirements to be met during its use (with examples);

— its usage.

e) Markings

— mass, if the mass of the object is more than 25 kg, this should be marked on the machine or the object;

— mass distribution if not equally distributed;

— positioning of the mark (on object, packaging);

— signs with prohibited applications should be provided;

— signs with precautions about shifting centre of gravity;

— defective packaging material.

f) Instruction

— what technical equipment must be used and when;

— description of proper working techniques and intended working practices;

— conditions for construction, transport and commissioning, use, decommissioning, disposal and dismantling;

— prohibited applications.

The instructions should be presented in a handbook.

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Annex A
(informative)

Populations characteristics and system design

A.1 Population characteristics

The table presented here applies to the general working population. This information is in accordance with measurements of maximum energetic capacity, subjective estimation of tolerability limits and objective measurements of physical capabilities[5)].

Table A.1 — Population percentages in relation to measurement criteria and the object mass

Options	Psychophysical data indicating tolerability capacity	Measurements of forces indicating limits	Measurements on the maximum metabolic ability limits
10 kg	99 % (F + M) 99 % F 99,9 % M	99 % (F + M) 99 % F 99,9 % M	99 % (F + M) 99 % F 99,9 % M
20 kg	95 % (F + M) 90 % F 99,9 % M	95 % (F + M) 90 % F 99,9 % M	95 % (F + M) 80 to 85 % F 99 % M
25 kg	85 % (F + M) 75 % F 99,9 % M	85 % (F + M) 72 to 75 % F 99,9 % M	85 % (F + M) 70 % F 99 % M
F: Female M: Male			

In addition to this table the following sub-populations have been identified as groups with an increased risk of sustaining injury: the young, the elderly, pregnant women and people with a history of back complaints. For pregnant women undertaking regularly lifting a value exceeding 5 kg is not advisable and for the young and elderly population this should not exceed 15 kg.

A.2 System design

The following interrelated aspects should be considered:

A.2.1 Avoid constrained postures

Thought should be given to the design/redesign of machines which lead to constrained working postures and monotonous work. In these circumstances discomfort and fatigue increases rapidly and muscular efficiency falls. Machinery should be designed to minimise, as far as possible, static postures.

A.2.2 Acceleration and movement accuracy

Acceleration places higher force requirements and strain on the body. Movement accuracy increases the time needed for manual handling and increases muscle effort. Accuracy of precise positioning should be supplied by the design.

5) Application of this information is limited and it is inappropriate to consider these values as safe limits. It will be the task of future research to confirm safe limits for the data contained in this standard.

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A.2.3 Minimise the discomfort, fatigue and stress on the operator

Research and experience in industry has shown that preventing discomfort, fatigue and stress during physically demanding work reduces ill-health and increases output. It is important to consider three factors:

a) physiological effort required;

b) the amount of work in constrained postures;

c) the large variation in individual susceptibility to fatigue.

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Annex B
(informative)

Recommended thermal comfort requirements

It is recommended to apply EN ISO 7730 for thermal comfort requirements.

The recommended limits for thermal comfort requirements during manual handling are 19 ° C to 26 °C; 30 % to 70 % humidity and air velocity ≤ 0,2 m/s.

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Annex C
(informative)

Risk assessment worksheets

The risk assessment model consists of three methods increasing in the level of complexity. The first method is a quick screening procedure to assess the task. Method 2 should be applied if the screening procedure indicates risks. This method takes account of additional risk factors. It is advisable to begin the risk assessment by applying method 1 (the most simple procedure) and use methods 2 or 3 only if the assumptions or operational situations identified in method 1 are not met.

Each method requires three steps to be carried out:

Step 1: Consider the reference mass

in relation to the intended user population (see Table C.1).

Step 2: Assess the risk factors

according to the worksheet.

Step 3: Identify the action required:

— no action, if risk level is tolerable;

— redesign, if the risk level is not tolerable or check that the risk is tolerable;

— use a more complex risk assessment method.

Table C.1 — Reference mass (M_{ref}) taking into consideration the intended user population[6)]

Field of application	M_{ref} [kg]	Percentage of			Population group	
		F and M	F_{emales}	M_{ales}		
Domestic use[a]	5	Data not available			Children and the elderly	Total population
	10	99	99	99	General domestic population	
Professional use (general)[b]	15	95	90	99	General working population, including the young and old	General working population
	25	85	70	90	Adult working population	
Professional use (exceptional)[c]	30 35 40	Data not available			Special working population	Special working population

[a] When designing a machine for domestic use, 10 kg should be used as a general reference mass in the risk assessment. If children and elderly are included in the intended user population, the reference mass should be lowered to 5 kg.

[b] When designing a machine for professional use, a reference mass of 25 kg should not be exceeded in general.

[c] While every effort should be made to avoid manual handling activities or reduce the risks to the lowest possible level, there may be exceptional circumstances where the reference mass might exceed 25 kg (e.g. where technological developments or interventions are not sufficiently advanced). Under these special conditions other measures have to be taken to control the risk according to EN 614-1 (e.g. technical aids, instructions and / or special training for the intended operator group).

6) Identical with Table 1.

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Worksheet 1

Risk assessment: Method 1 - Screening by means of critical values

EN 1005 Safety of machinery - Human physical performance -
Part 2: Manual handling of machinery and component parts of machinery

This method provides a quick screening procedure to identify whether the handling operation represent a risk to the operator(s). Step 2 requires one of three critical operational situations (cases 1 to 3) to be selected. The limiting condition is that all assumptions for handling operations are fulfilled.

Step 1: Consider the reference mass

Identify the intended user population and select the reference mass (M_{ref}) according to the intended user population (Table C.1)

Step 2: Carry out the risk assessment

Please tick the following criteria for the handling operation, if met:

- ❑ two-handed operation only
- ❑ unrestricted standing posture and movements
- ❑ handling by one person only
- ❑ smooth lifting
- ❑ good coupling between the hands and the objects handled
- ❑ good coupling between the feet and floor
- ❑ manual handling activities, other than lifting, are minimal
- ❑ the objects to be lifted are not cold, hot or contaminated
- ❑ moderate ambient thermal environment

If one or more of these criteria are not met, refer to method 2.
If all criteria are met, then select one of the following critical variables. These apply to a work shift of 8 h or less.

Case 1 *critical mass*

- ❑ the load handled does not exceed 70 % of the reference mass selected from Table C.1
- ❑ vertical displacement of the load is less or equal 25 cm and between hip and shoulder height
- ❑ the trunk is upright and not rotated
- ❑ the load is kept close to the body
- ❑ the frequency of lifts is equal to or less than 0,00333 Hz (1 lift every 5 min)

Case 2 *critical vertical mass displacement*

- ❑ the load handled does not exceed 60 % of the reference mass selected from Table C.1
- ❑ vertical displacement of the load is not above shoulder height or below knee height
- ❑ the trunk is upright and not rotated
- ❑ the load is kept close to the body
- ❑ the frequency of lifts is equal to or less than 0,00333 Hz (1 lift every 5 min)

Case 3 *critical frequency*

- ❑ the load handled does not exceed 30 % of the reference mass selected from Table C.1
- ❑ vertical displacement of the load is less or equal 25 cm and between hip and shoulder height
- ❑ the frequency of lifts is equal or less than 0,08 Hz (5 lifts every min)
- ❑ the trunk is upright and not rotated
- ❑ the load is kept close to the body

Or

- ❑ the load handled does not exceed 50 % of the reference mass selected from Table C.1
- ❑ vertical displacement of the load is less or equal 25 cm and between hip and shoulder height
- ❑ the frequency of lifts is equal or less than 0,04 Hz (2,5 lifts every min)
- ❑ the trunk is upright and not rotated
- ❑ the load is kept close to the body

Step 3: Select the action required

If the design fits one of the operational situations (cases 1 to 3) described above, the risk assessment has been carried out successfully.
If none of the operational situations are satisfied, or any of the criteria specified in step 2 are not met, either

⇒ consider modifying or redesign the machinery or
⇒ use a more detailed risk assessment procedure to identify critical risk factors (method 2).

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Worksheet 2a

Risk assessment: Method 2 - Estimation by tables

EN 1005 Safety of machinery - Human physical performance -
Part 2: Manual handling of machinery and component parts of machinery

Step 1: Consider the reference mass

Identify the intended user population and select the reference mass (M_{ref}) according to the intended user population (Table C.1)

Step 2: Carry out the risk assessment

Please indicate (tick), whether the handling operation meets the following criteria:

- ❑ two-handed operation only
- ❑ unrestricted standing posture and movements
- ❑ handling by one person only
- ❑ smooth lifting
- ❑ good coupling between the feet and floor
- ❑ manual handling activities, other than lifting, are minimal
- ❑ the objects to be handled are not cold, hot or contaminated
- ❑ moderate ambient thermal environment

If one or more of these criteria are not met, refer to method 3.

If all criteria are met, then determine the level of risk by:

1. calculating the recommended mass limit (R_{ML2}) using the multipliers provided in Table C.1
2. calculating the risk index (R_I) as follows:

$$\text{risk index } (R_I) = \frac{\text{actual mass}}{R_{ML}} = \frac{\quad}{\quad} \begin{matrix}[\text{kg}]\\[\text{kg}]\end{matrix}$$

Step 3: Select the action required

- $\boldsymbol{R_I \leq 0{,}85}$ the risk may be regarded as tolerable.
- $\boldsymbol{0{,}85 < R_I < 1{,}0}$ significant risk exists. It is recommended that:
  - ⇒ Method 3 is applied in order to identify how the risk may be reduced or
  - ⇒ the machinery be either redesigned or
  - ⇒ ensure, that the risk is tolerable.
- $\boldsymbol{R_I \geq 1{,}0}$ redesign is necessary. The design can be improved by changing the situations that lead to low multipliers.

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Worksheet 2b

Risk assessment: Method 2 - Estimation by tables

EN 1005 Safety of machinery - Human physical performance -
Part 2: Manual handling of machinery and component parts of machinery

Table C.2 — Calculation of the recommended mass limit (R_{ML2})

R_{ML2}

Reference mass (M_{ref}) =

Reference mass [kg] (see Table C.1)	M_{ref}

Vertical multiplier (V_M) x

Vertical location [cm]	0	25	50	75	100	130	> 175	V_M
factor	0,78	0,85	0,93	1,00	0,93	0,84	0,00	

Distance multiplier (D_M) x

Vertical displacement [cm]	25	30	40	50	70	100	> 175	D_M
factor	1,00	0,97	0,93	0,91	0,88	0,87	0,00	

Horizontal multiplier (H_M) x

Horizontal location [cm]	25	30	40	50	55	60	> 63	H_M
factor	1,00	0,83	0,63	0,50	0,45	0,42	0,00	

Asymmetric multiplier (A_M) x

angle of asymmetry [°]	0	30	60	90	120	135	> 135	A_M
factor	1,00	0,90	0,81	0,71	0,62	0,57	0,00	

Coupling multiplier (C_M) x

quality of grip	good	fair	poor	C_M
description	load length ≤ 40 cm; load height ≤ 30 cm; good handles or hand-hold cut-outs. Easy to handle loose parts and objects with wrap around grasp and without excessive wrist deviation.	load length ≤40 cm; load height ≤ 30 cm; and poor handles or hand-hold cut-outs or 90° finger flexion. Easy to handle loose parts and objects with 90° finger flexion and without excessive wrist deviation.	load length > 40 cm or; load height > 30 cm; or difficult to handle parts or sagging objects or asymmetric center of mass or unstable contents or hard to grasp object or use of gloves.	
Factor	1,00	0,95	0,90	

Frequency multiplier (F_M) dependent from work duration (d) x

		frequency							
	[Hz]	0,0033	0,0166	0,0666	0,1000	0,1500	0,2000	> 0,2500	F_M
	[lifts / min]	0,2	1	4	6	9	12	> 15	
work duration (d)	d ≤ 1 h	1,00	0,94	0,84	0,75	0,52	0,37	0,00	
	1 h < d ≤ 2 h	0,95	0,88	0,72	0,50	0,30	0,00	0,00	
	2 h< d ≤ 8 h	0,85	0,75	0,45	0,27	0,00	0,00	0,00	

=

$R_{ML2} = M_{ref} \times V_M \times D_M \times H_M \times A_M \times C_M \times F_M$	=	[kg]

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Worksheet 3a

Risk assessment: Method 3 - Calculation by formula

EN 1005 Safety of machinery - Human physical performance -
Part 2: Manual handling of machinery and component parts of machinery

Step 1: Consider the reference mass

Identify the intended user population and select the reference mass (M_{ref}) according to the intended user population (Table C.1)

Step 2: Carry out the risk assessment

Please identify (tick), whether the handling operation meets the following criteria:

- ❑ unrestricted standing posture and movements;
- ❑ smooth lifting;
- ❑ good coupling between the feet and floor;
- ❑ the objects to be handled are not cold, hot or contaminated;
- ❑ moderate ambient thermal environment.

If one or more of the criteria are not met, consider ways of meeting each of the criteria, refer to clause 4 of this standard.
If all criteria are met, calculate the recommended mass limit (R_{ML}).

Case 1

If the recommended mass limit (R_{ML2}) is already known (calculated during method 2)

then calculate the recommended mass limit (R_{ML}) as follows:

$\boldsymbol{R_{ML} = R_{ML2} \times O_M \times P_M \times A_T}$ **[kg]**

where:

O_M one handed operation	if true O_M **= 0,6**	otherwise O_M **= 1,0**
P_M two person operation	if true P_M **= 0,85**	otherwise P_M **= 1,0**
A_T additional physically demanding tasks	if true A_T **= 0,8**	otherwise A_T **= 1,0**

Case 2

If the recommended mass limit (R_{ML}) has <u>not</u> been calculated

then calculate the recommended mass limit (R_{ML}) as follows:

$\boldsymbol{R_{ML} = M_{ref} \times V_M \times D_M \times H_M \times A_M \times C_M \times F_M \times O_M \times P_M \times A_T}$

the following definitions apply:

$V_M = 1 - 0{,}003\,\lvert V - 75\rvert$	if $V < 0$ cm, $V_M = 0{,}78$	if $V > 175$ cm, $V_M = 0$
$D_M = 0{,}82 + 4{,}5/D$	if $D < 25$ cm, $D_M = 1$	if $D > 175$ cm, $D_M = 0$
$A_M = 1 - (0{,}0032A)$		if $A > 135°$, $A_M = 0$
$H_M = 25/H$	if $H < 25$ cm, $H_M = 1$	if $H > 63$ cm, $H_M = 0$

M_{ref}	the reference mass from Table C.1, in kg
V	vertical location of the load, in cm
D	vertical displacement of the load, in cm
H	horizontal location of the load, in cm
A	angle of asymmetry, in degree
C_M	coupling multiplier from Table C.1
F_M	frequency multiplier from Table C.2

O_M one handed operation	if true O_M **= 0,6**	otherwise O_M **= 1,0**
P_M two person operation	if true P_M **= 0,85**	otherwise P_M **= 1,0**
A_T additional physically demanding tasks	if true A_T **= 0,8**	otherwise A_T **= 1,0**

calculate the risk index (R_I) as follows:

$$\text{risk index } (R_I) = \frac{\text{actual mass}}{R_{ML}} = \frac{\quad}{} \; \frac{[\text{kg}]}{[\text{kg}]}$$

Step 3: Select the action required

- $\boldsymbol{R_I \leq 0{,}85}$ — the risk may be regarded as tolerable.
- $\boldsymbol{0{,}85 < R_I < 1{,}0}$ — significant risk exists. It is recommended to:
  - ⇒ redesign the machinery or
  - ⇒ ensure, that the risk is tolerable.
- $\boldsymbol{R_I \geq 1{,}0}$ — redesign is necessary. The design can be improved by changing the situations that lead to low multipliers.

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Worksheet 3b

Risk assessment: Method 3 - Calculation by formula

EN 1005 Safety of machinery - Human physical performance -
Part 2: Manual handling of machinery and component parts of machinery

Table C.3 — Coupling multiplier (CM)

quality of grip	good	fair	poor
description	load length ≤ 40 cm; load height ≤ 30 cm; good handles or hand-hold cut-outs. Easy to handle loose parts and objects with wrap around grasp and without excessive wrist deviation.	load length ≤ 40 cm; load height ≤ 30 cm; and poor handles or hand-hold cut-outs or 90° finger flexion. Easy to handle loose parts and objects with 90° finger flexion and without excessive wrist deviation.	load length > 40 cm or; load height > 30 cm; or difficult to handle parts or sagging objects or asymmetric center of mass or unstable contents or hard to grasp object or use of gloves.
factor	1,00	0,95	0,90

Table C.4 — Frequency multiplier (FM)

frequency		work duration *d*					
		2 h < *d* ≤ 8 h		1 h < *d* ≤ 2 h		*d* ≤ 1 h	
[Hz]	[Lifts/minute]	*V* < 75 cm	*V* ≥ 75 cm	*V* < 75 cm	*V* ≥ 75 cm	*V* < 75 cm	*V* ≥ 75 cm
≤ 0,00333	≤ 0,2	0,85	0,85	0,95	0,95	1,00	1,00
0,00833	0,5	0,81	0,81	0,92	0,92	0,97	0,97
0,01666	1	0,75	0,75	0,88	0,88	0,94	0,94
0,03333	2	0,65	0,65	0,84	0,84	0,91	0,91
0,05000	3	0,55	0,55	0,79	0,79	0,88	0,88
0,06666	4	0,45	0,45	0,72	0,72	0,84	0,84
0,08333	5	0,35	0,35	0,60	0,60	0,80	0,80
0,10000	6	0,27	0,27	0,50	0,50	0,75	0,75
0,11666	7	0,22	0,22	0,42	0,42	0,70	0,70
0,13333	8	0,18	0,18	0,35	0,35	0,60	0,60
0,15000	9	0,00	0,15	0,30	0,30	0,52	0,52
0,16666	10	0,00	0,13	0,26	0,26	0,45	0,45
0,18333	11	0,00	0,00	0,00	0,23	0,41	0,41
0,20000	12	0,00	0,00	0,00	0,21	0,37	0,37
0,21666	13	0,00	0,00	0,00	0,00	0,00	0,34
0,23333	14	0,00	0,00	0,00	0,00	0,00	0,31
0,25000	15	0,00	0,00	0,00	0,00	0,00	0,28
>0,2500	>15	0,00	0,00	0,00	0,00	0,00	0,00

V is the vertical location

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Annex ZA
(informative)

A1 Relationship between this European Standard and the Essential Requirements of EU Directive 98/37/EC, amended by 98/79/EC

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide a means of conforming to Essential Requirements of the New Approach Directive 98/37/EC on machinery, amended by 98/79/EC.

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the relevant Essential Requirements of that Directive and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Directive 98/37/EC, amended by 98/79/EC

Clause(s)/sub-clause(s) of this EN	Essential Requirements (ERs) of Directive 98/37/EC, amended by 98/79/EC	Qualifying remarks/Notes
All clauses	Annex I: 1.1.2 d), 1.1.5	-

WARNING — Other requirements and other EU Directives may be applicable to the product(s) falling within the scope of this standard. A1

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Annex ZB
(informative)

A1 Relationship between this European Standard and the Essential Requirements of EU Directive 2006/42/EC

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide a means of conforming to Essential Requirements of the New Approach Directive 2006/42/EC on machinery.

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the normative clauses of this standard given in Table ZB.1 confers, within the limits of the scope of this standard, a presumption of conformity with the relevant Essential Requirements of that Directive and associated EFTA regulations.

Table ZB.1 — Correspondence between this European Standard and Directive 2006/42/EC

Clause(s)/sub-clause(s) of this EN	Essential Requirements (ERs) of Directive 2006/42/EC	Qualifying remarks/Notes
All clauses	Annex I: 1.1.5, 1.1.6	-

WARNING — Other requirements and other EU Directives may be applicable to the product(s) falling within the scope of this standard. A1

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